BAO CAO THUC NGHIEM LAB01

Run ID: 20261105_003135_11290d
Trang thai: SUCCESS
Thoi gian bat dau: 2026-11-05T00:31:35+07:00
Thoi gian ket thuc: 2026-11-05T00:32:23+07:00

1. TIEN TRINH THUC HIEN

STT	Buoc thuc hien	Trang thai
1	EDA	SUCCESS
2	PREPROCESSING	SUCCESS
3	TRAINING - Logistic Regression	SUCCESS
4	TRAINING - Naive Bayes	SUCCESS
5	TRAINING - SVM	SUCCESS
6	TRAINING	SUCCESS
7	EVALUATION - Logistic Regression - Train	SUCCESS
8	EVALUATION - Logistic Regression - Test	SUCCESS
9	EVALUATION - Naive Bayes - Train	SUCCESS
10	EVALUATION - Naive Bayes - Test	SUCCESS
11	EVALUATION - SVM - Train	SUCCESS
12	EVALUATION - SVM - Test	SUCCESS
13	EVALUATION	SUCCESS
14	PIPELINE	SUCCESS

2. KET QUA DANH GIA MO HINH

Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
Logistic Regression	Train	0.9535472972972973	0.9007832898172323	0.9530386740331491	0.9261744966442953	0.9530386740331491	0.046228710462287104	0.9716361438211613	3136	152	68	1380
Logistic Regression	Test	0.9315878378378378	0.8469135802469135	0.9475138121546961	0.8943937418513689	0.9475138121546961	0.075425790754257915	0.9596792622763506	760	62	19	343
Naive Bayes	Train	0.9961993243243243	0.9897260273972602	0.9979281767955801	0.9938101788170564	0.9979281767955801	0.004562043795620438	0.9999590424244869	3273	15	3	1445
Naive Bayes	Test	0.9729729729729729	0.9634831460674157	0.9475138121546961	0.9554317548746517	0.9475138121546961	0.015815085158150853	0.9935476065653103	809	13	19	343
SVM	Train	0.7193834459459459	0.8888888888888888	0.09392265193370165	0.16989381636477202	0.09392265193370165	0.005170316301703163	0.9747585813472059	3271	17	1312	136

Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
SVM	Test	0.71875	0.8918918918918919	0.09116022099447514	0.16541353383458646	0.09116022099447514	0.004866180048661801	0.9647269158903631	818	4	329	33

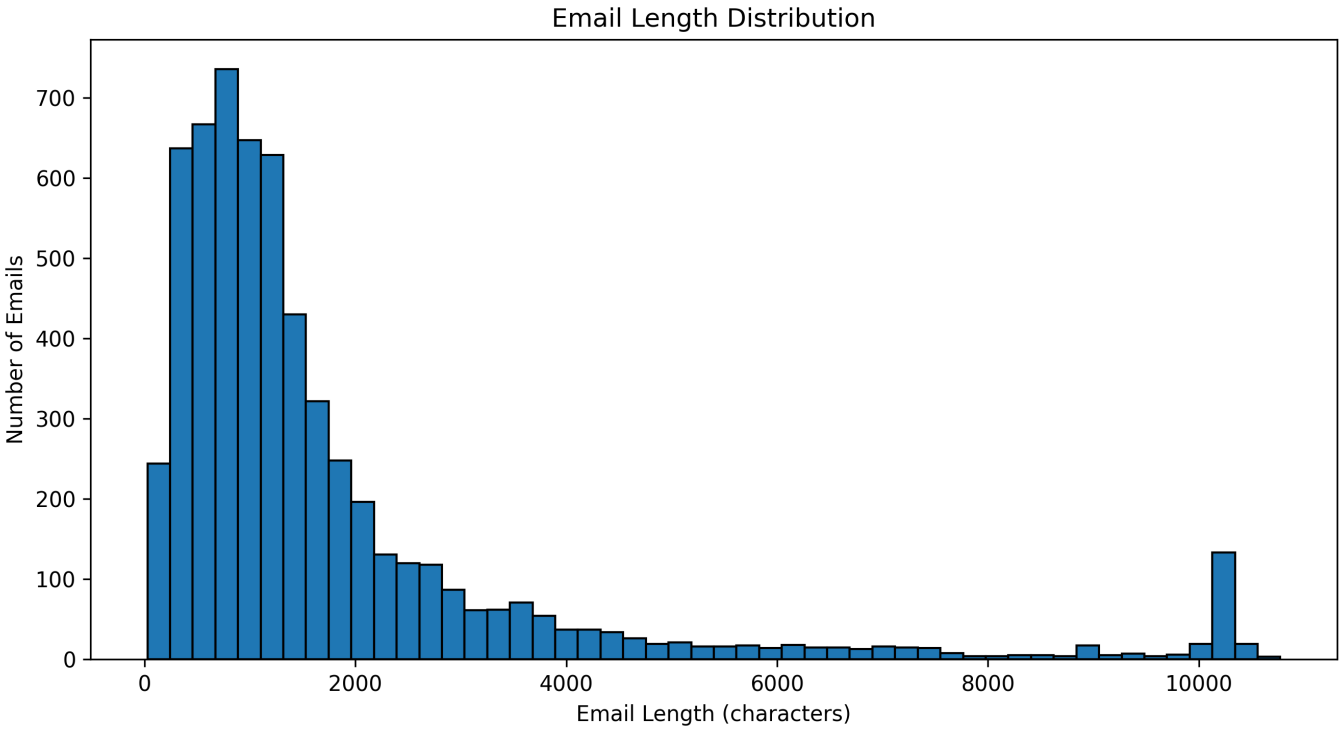
3. MO HINH TOT NHAT

Mo hinh: Naive Bayes
F1-score: 0.9554317548746517

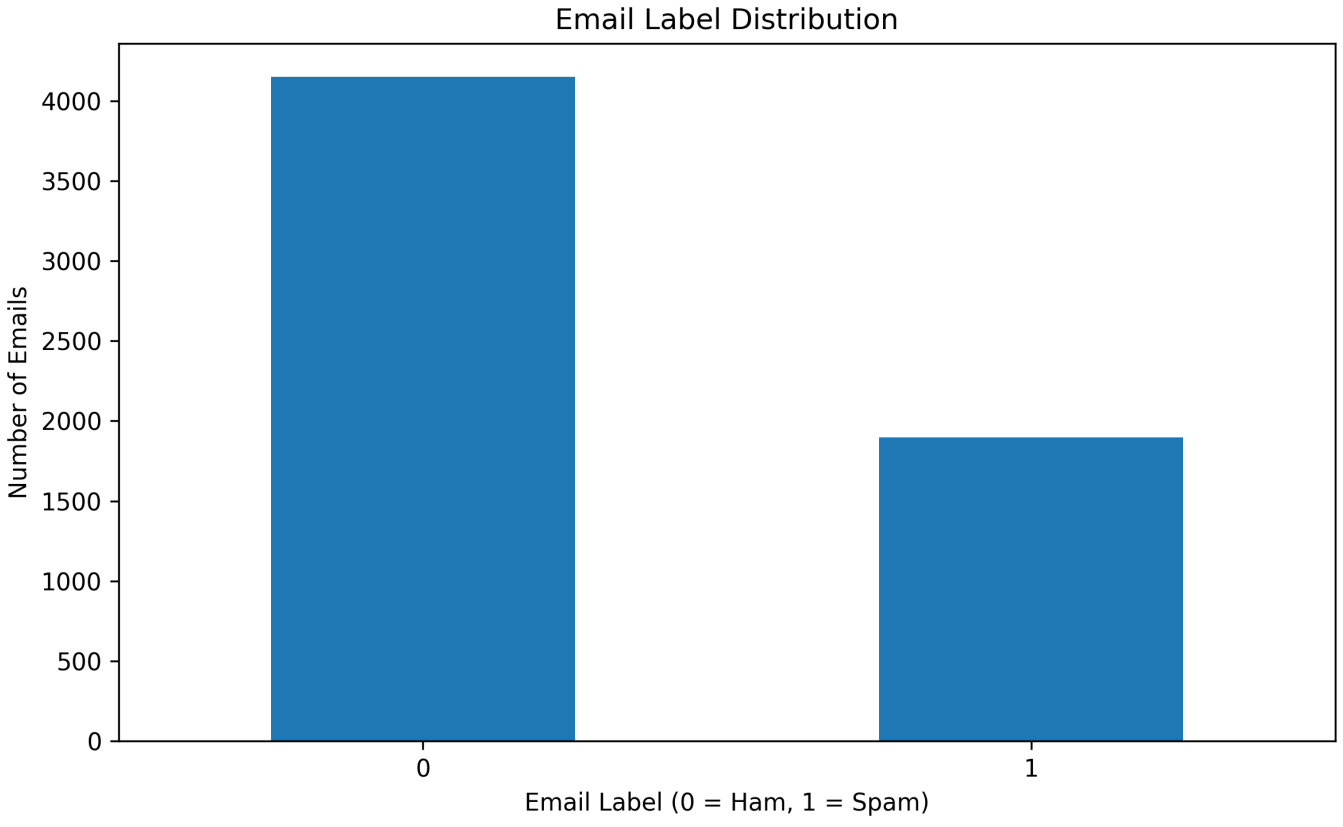
4. BIEU DO KET QUA

EDA

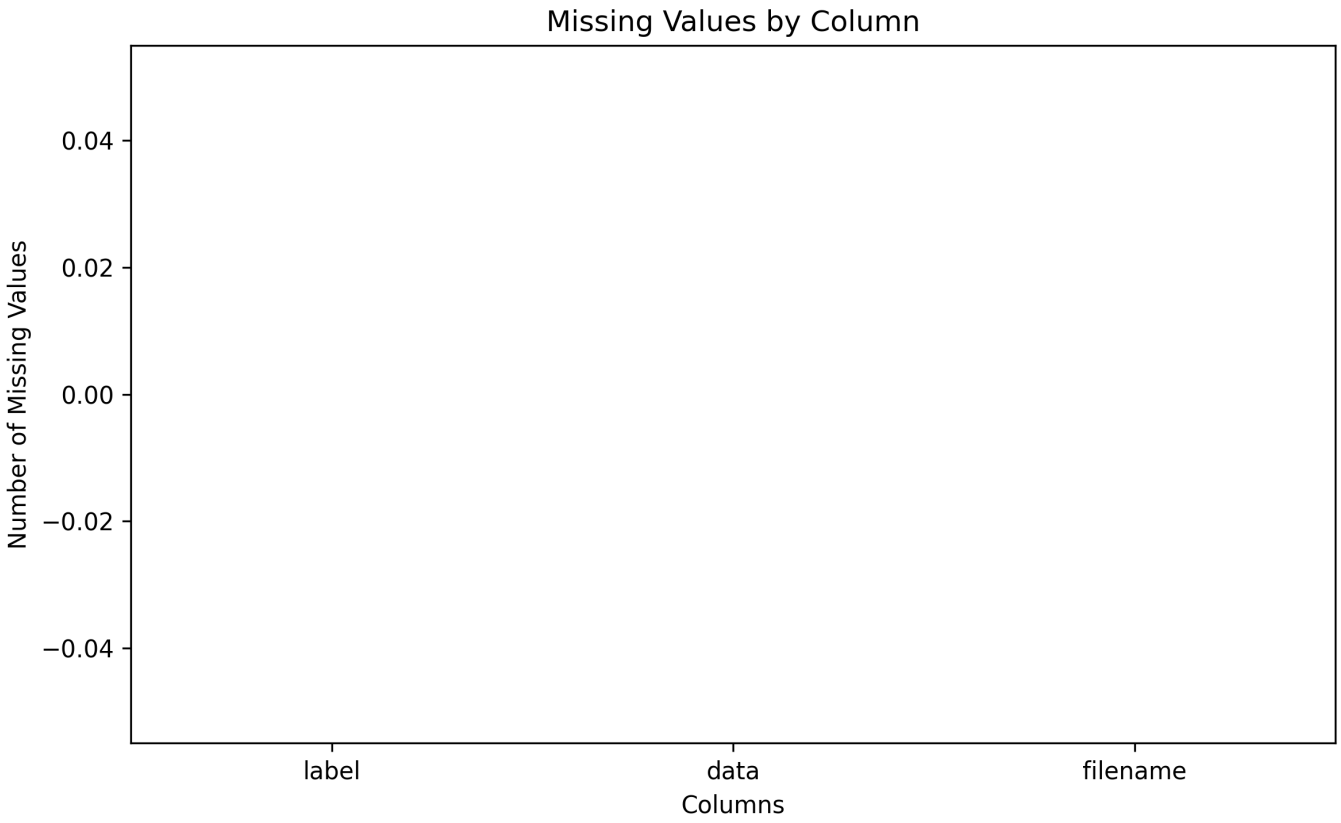
email_length_distribution



label_distribution

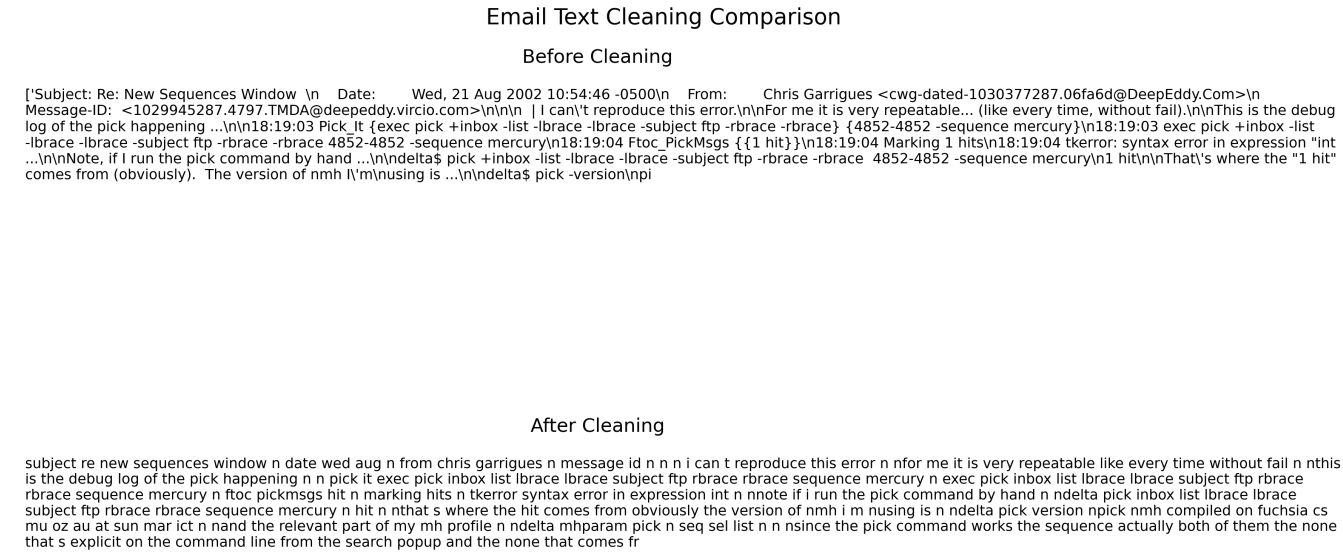


missing_values

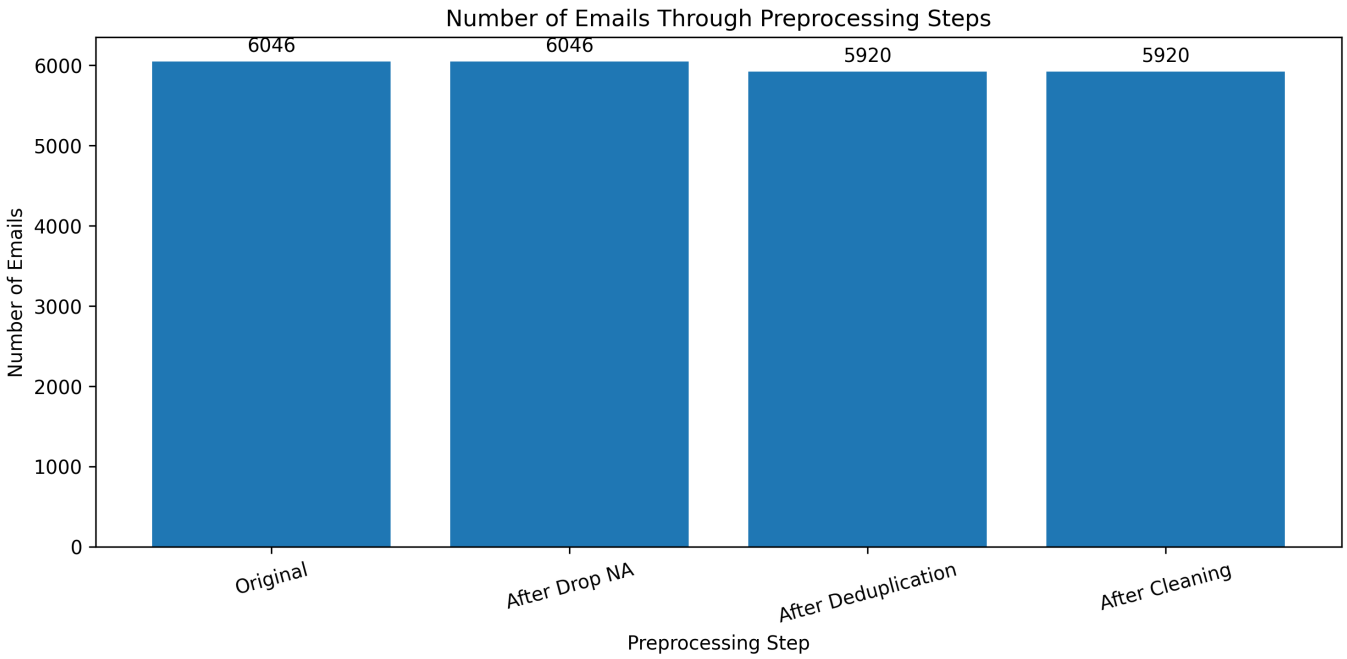


PREPROCESSING

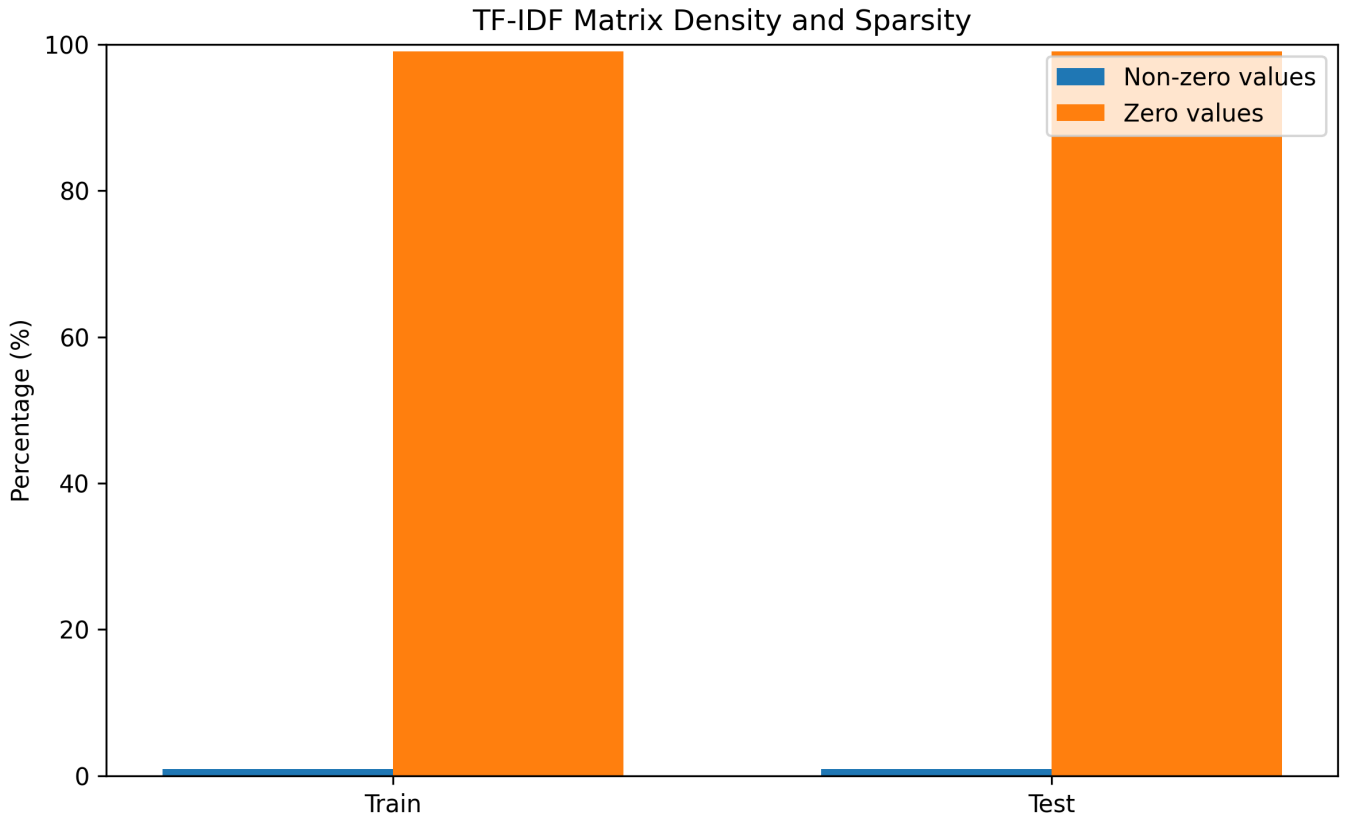
cleaning_comparison



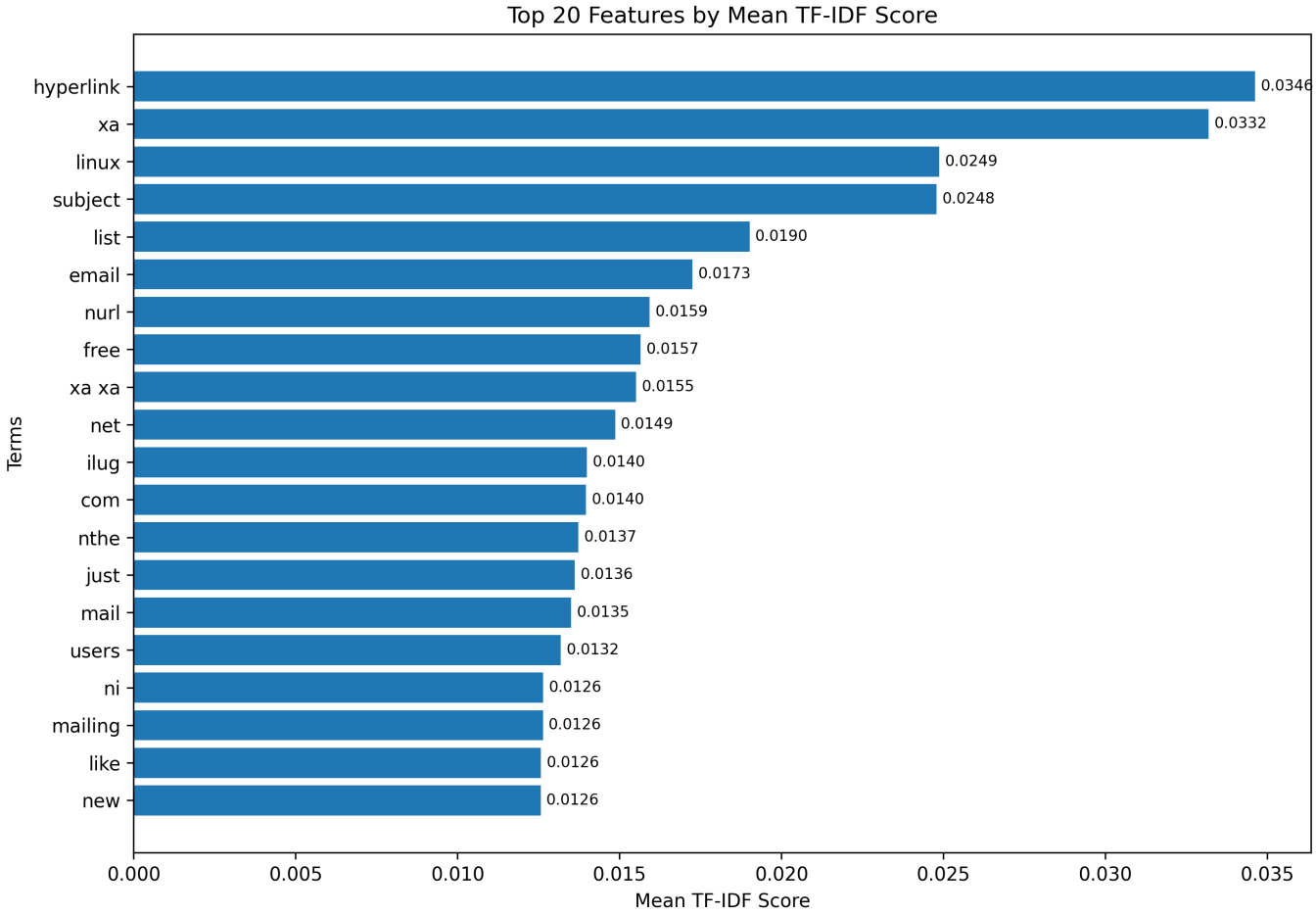
preprocessing_steps



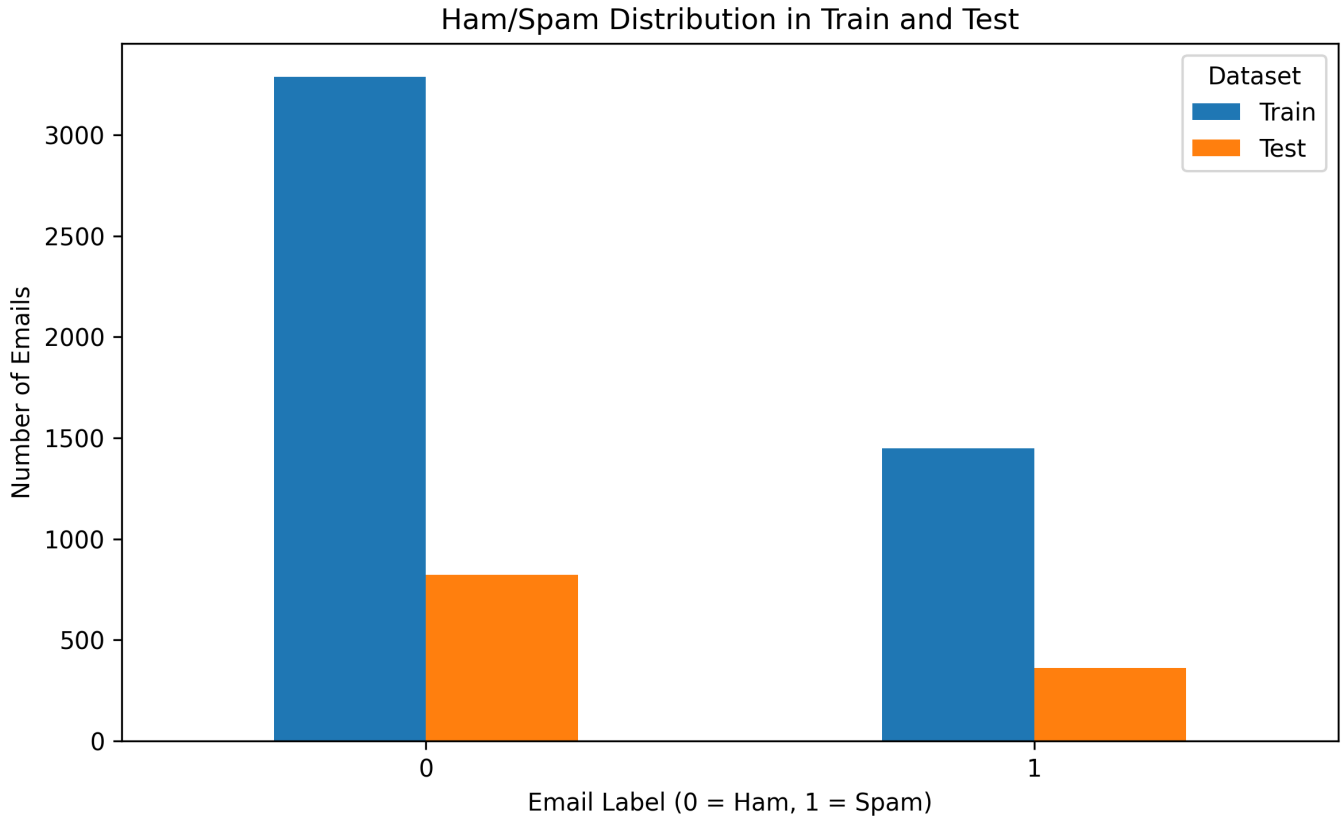
tfidf_sparsity



tfidf_top_features



train_test_distribution



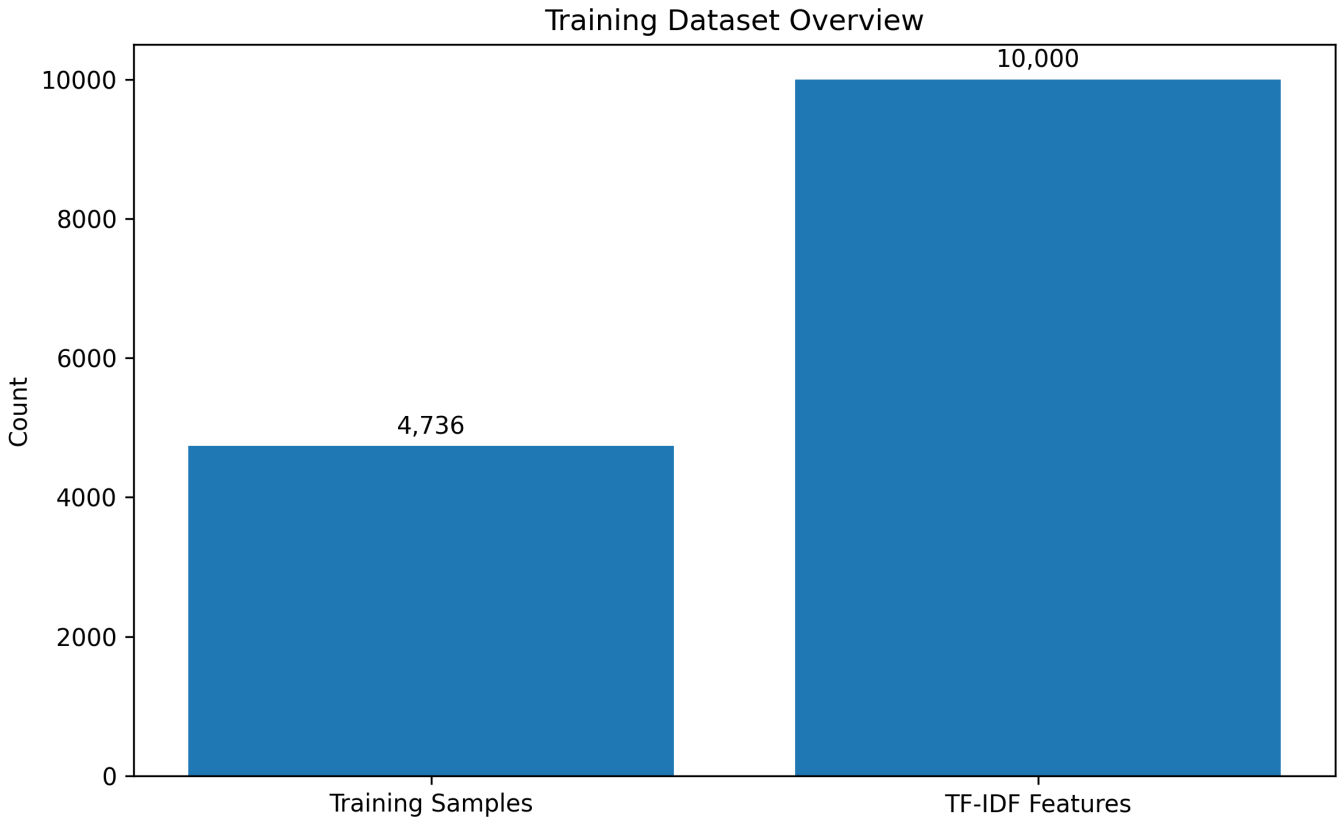
TRAINING

training_configuration

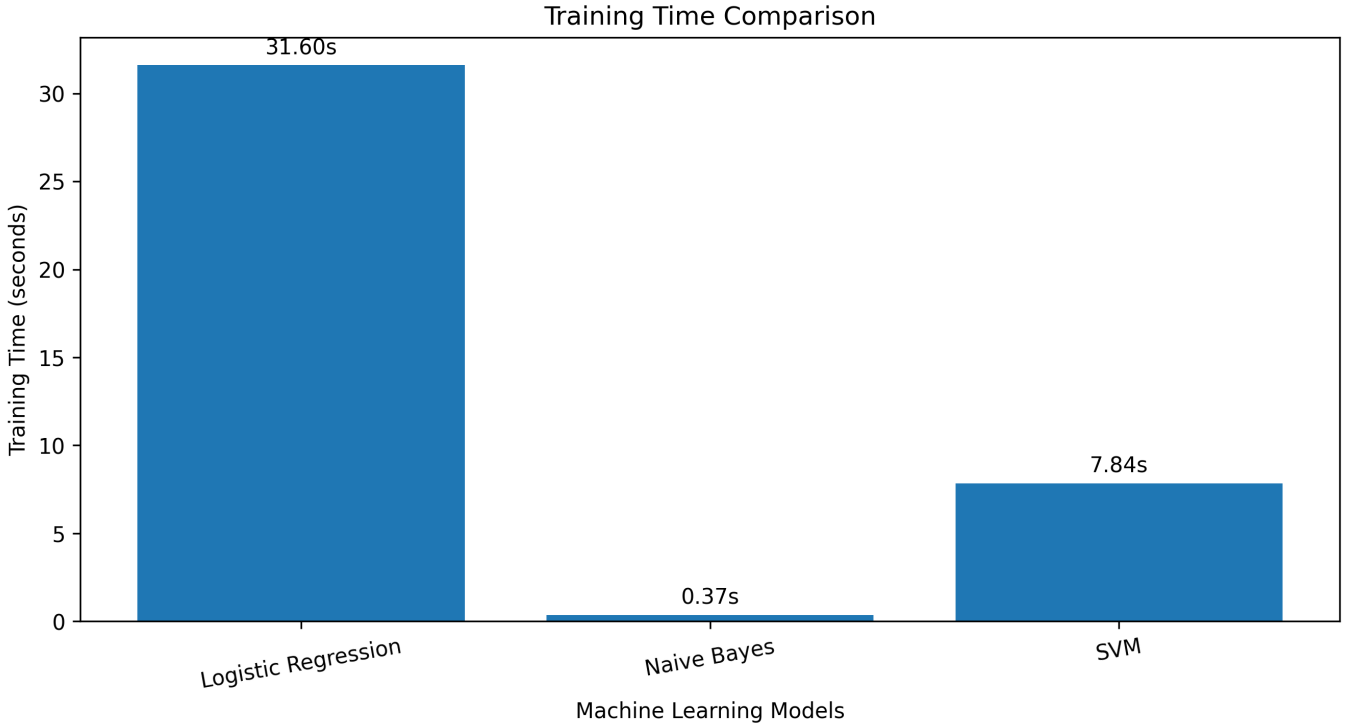
Model Training Configuration

Model	Learning Rate	Iterations	Lambda
Logistic Regression	0.1	1000	N/A
Naive Bayes	N/A	N/A	N/A
SVM	0.01	100	0.01

training_dataset_overview

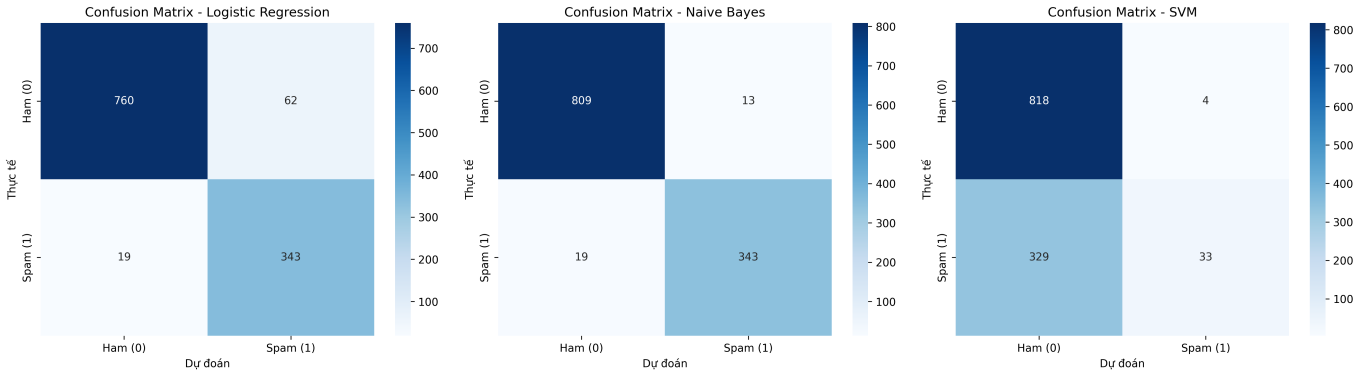


training_time_comparison

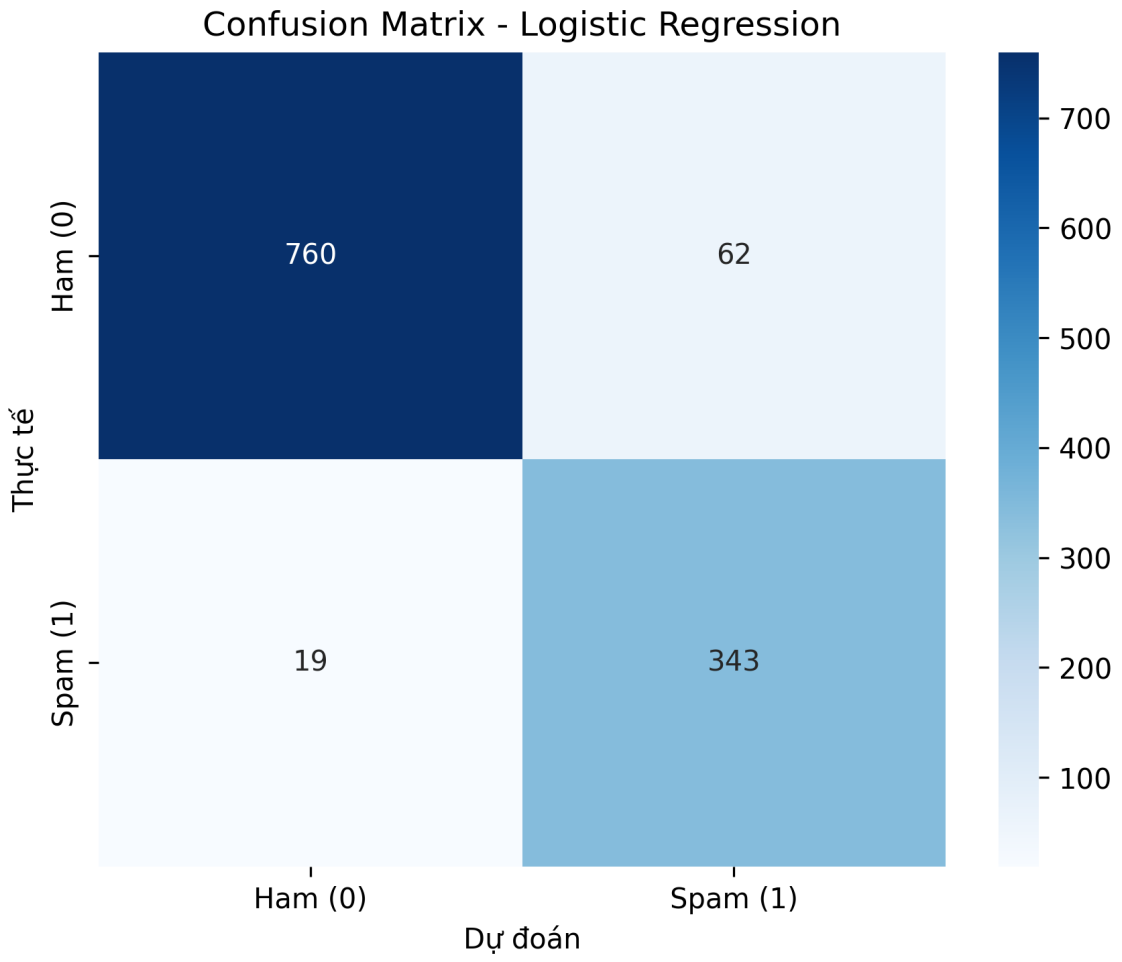


EVALUATION

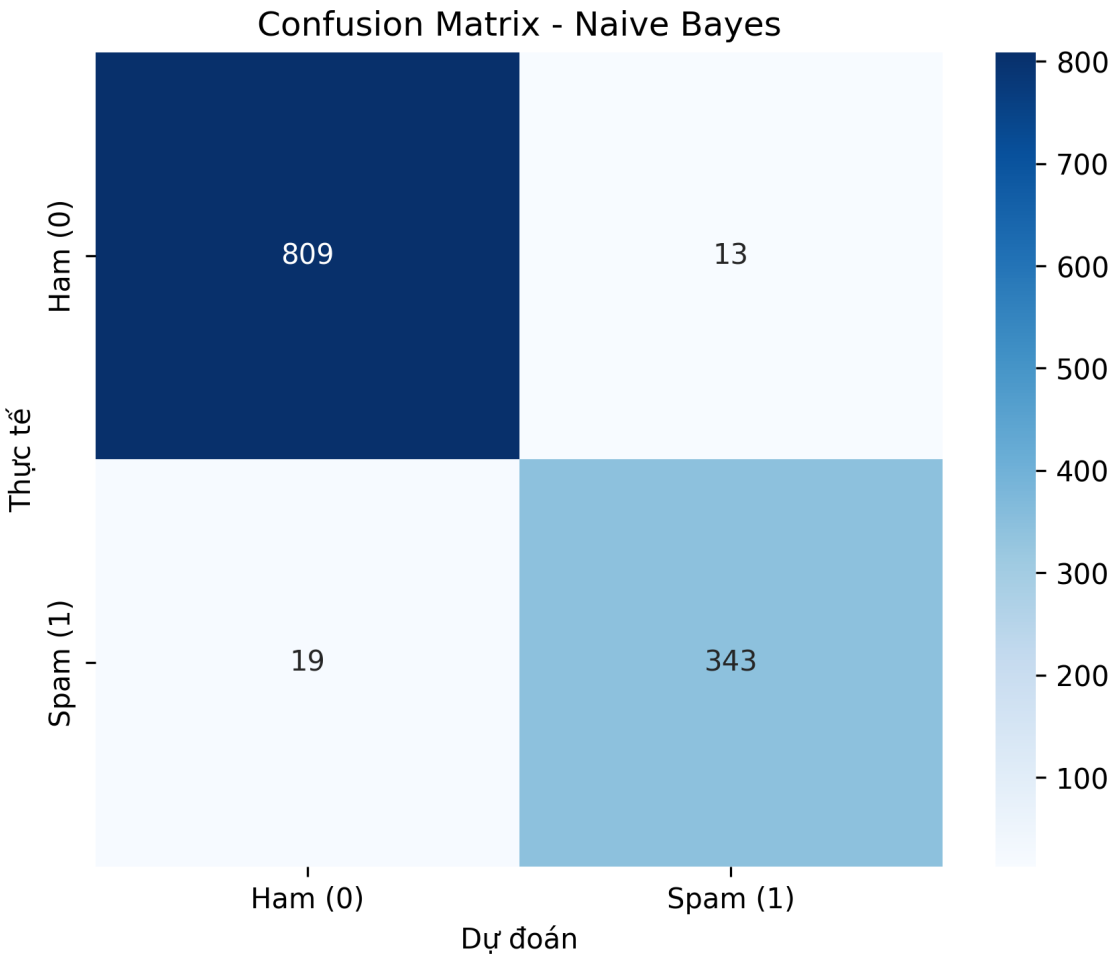
confusion_matrix_all



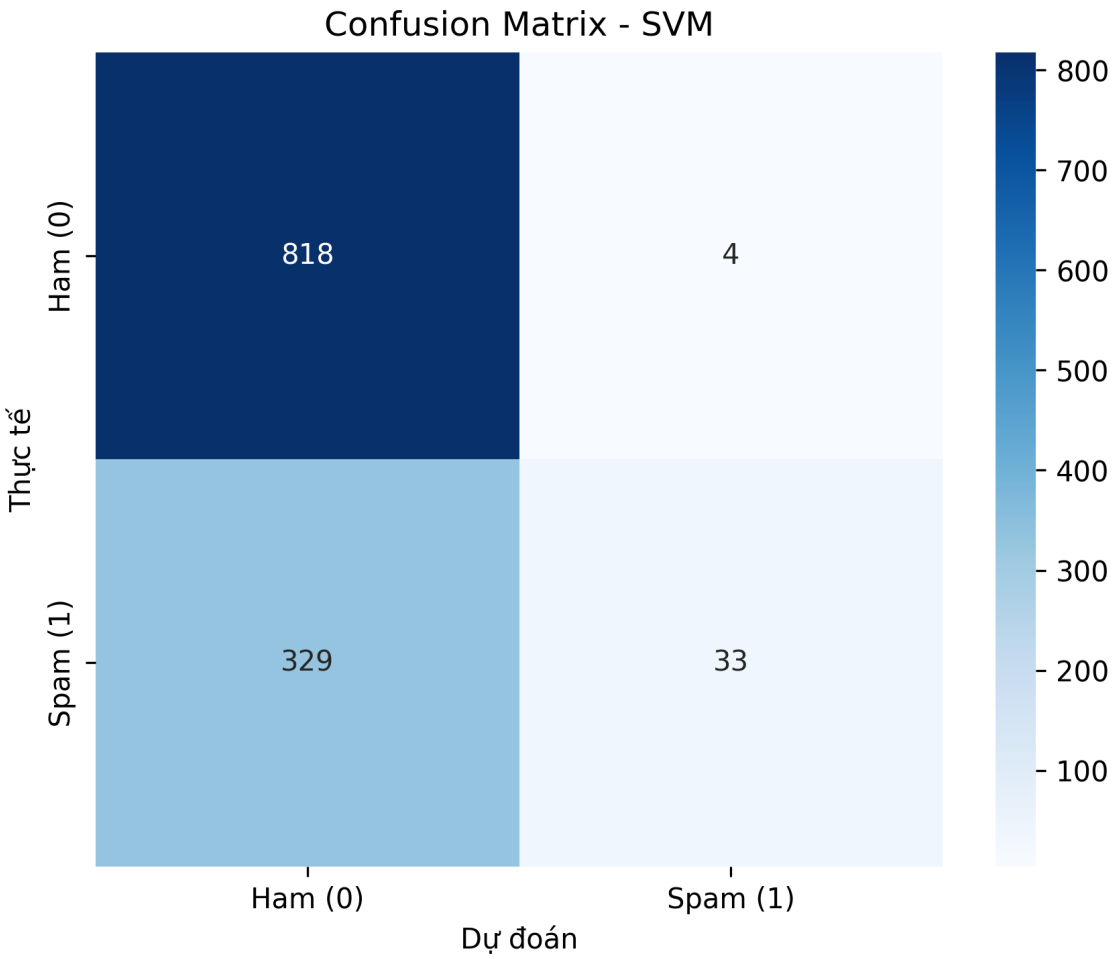
confusion_matrix_lr



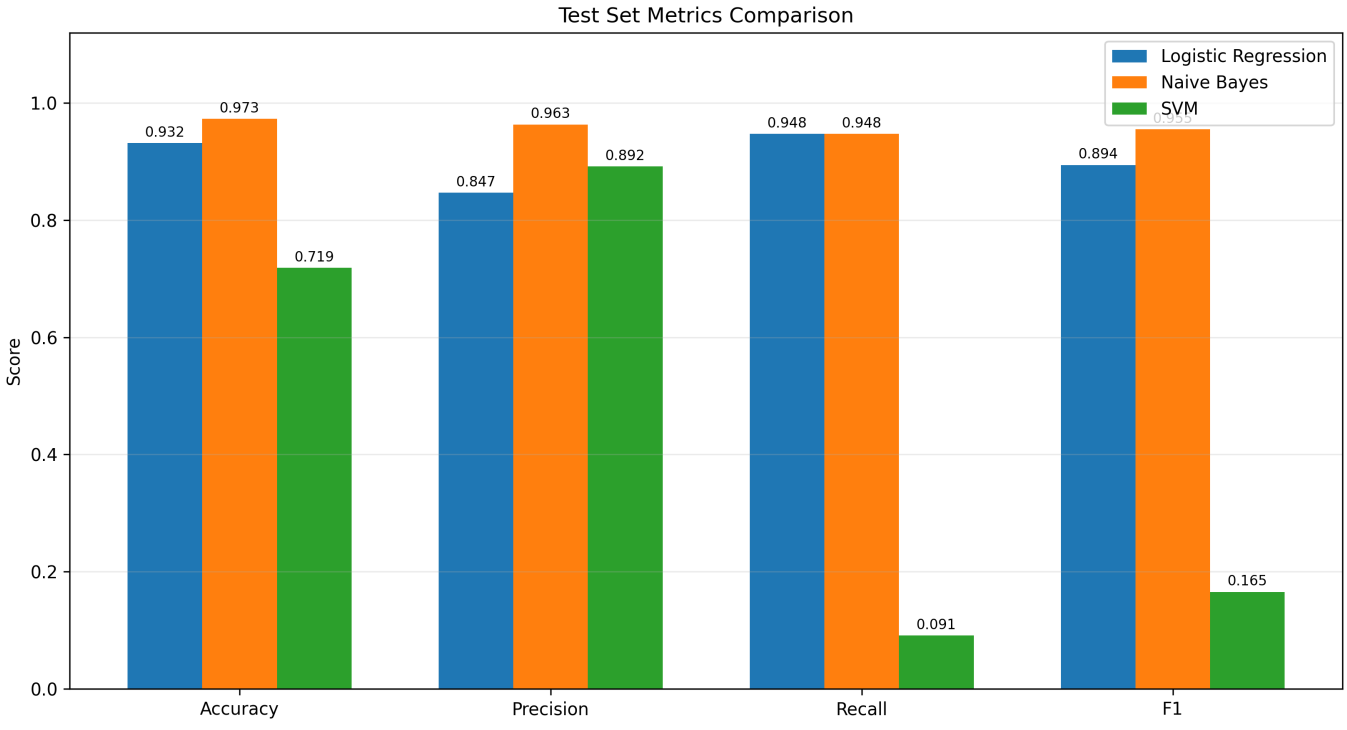
confusion_matrix_nb



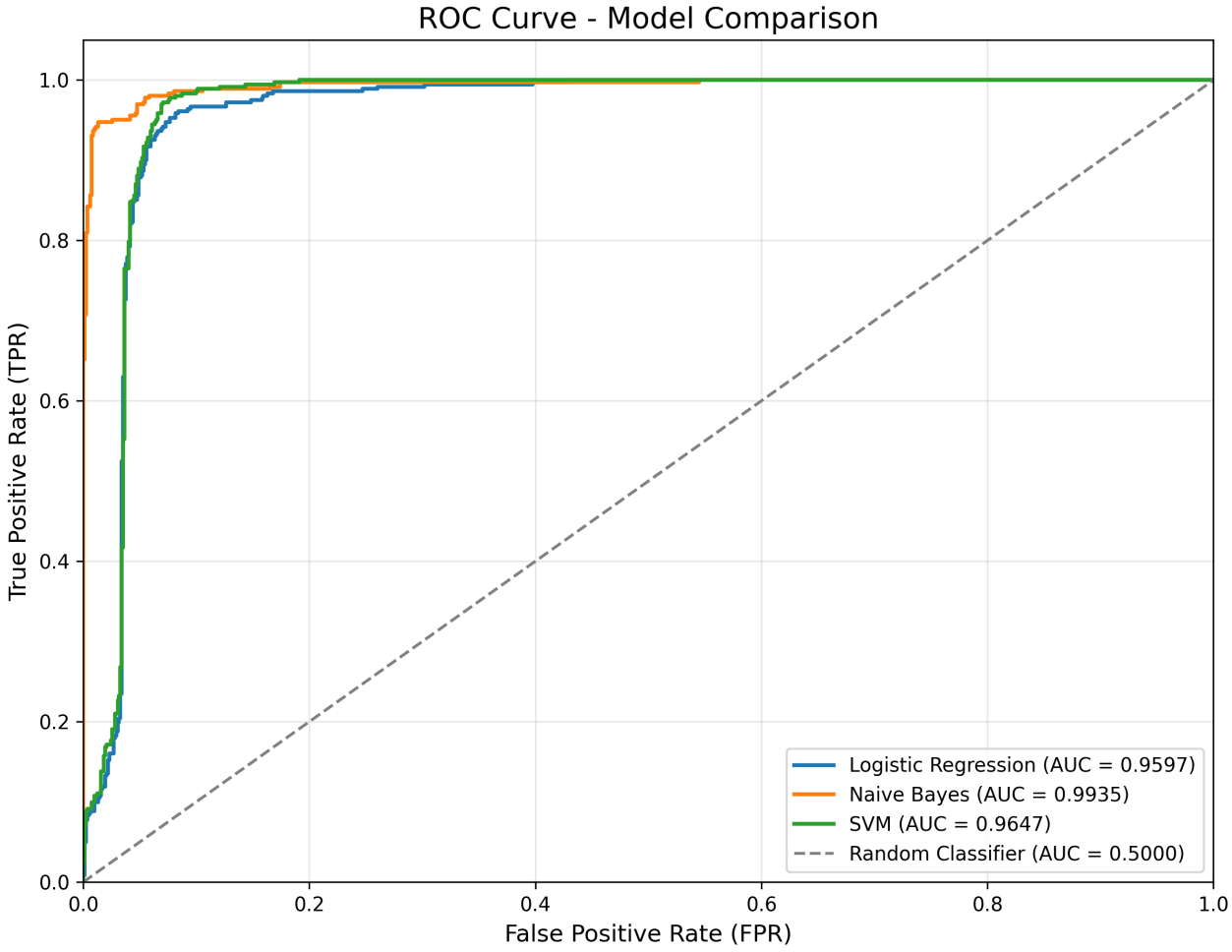
confusion_matrix_svm



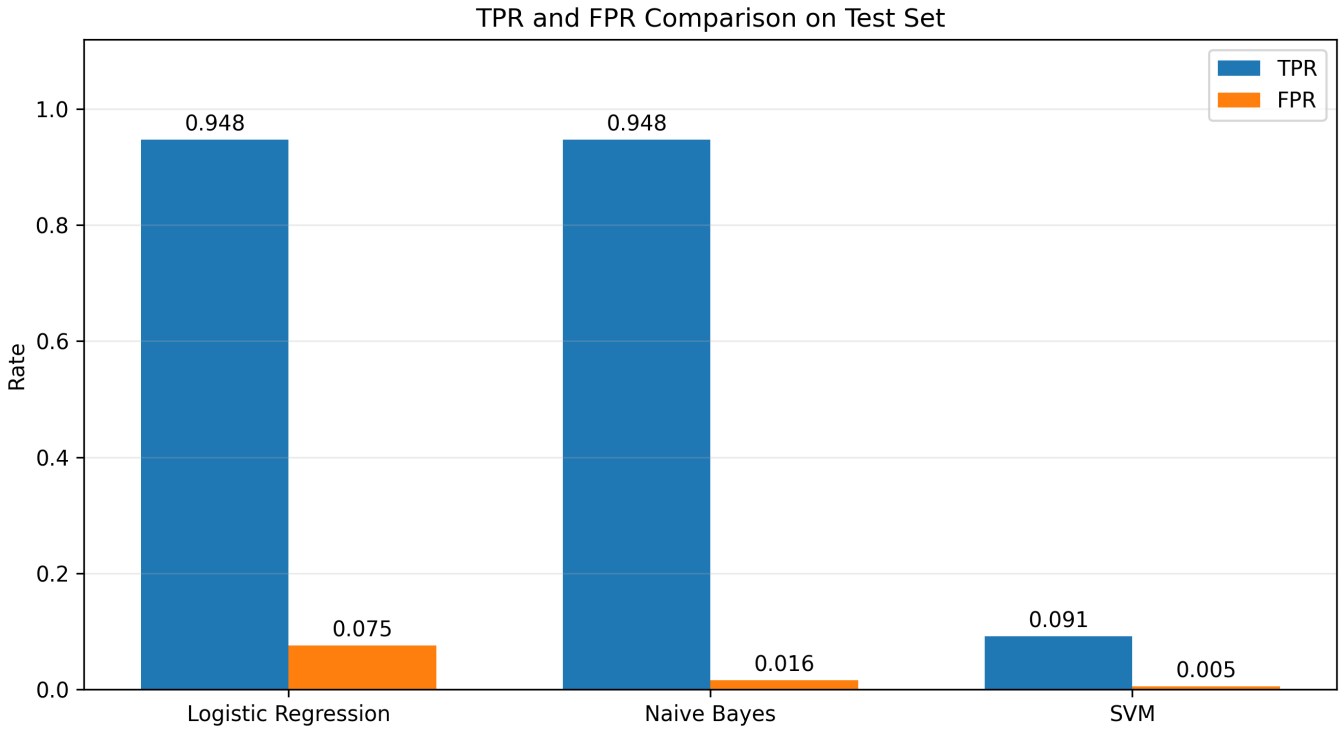
metrics_comparison



roc_curve_all



tpr_fpr_comparison



5. NHAT KY THUC THI

[2026-11-05T00:31:35+07:00] [START] [PIPELINE] Bat dau quy trinh may hoc.

[2026-11-05T00:31:35+07:00] [START] [EDA] Bat dau buoc EDA.

[2026-11-05T00:31:35+07:00] [START] [EDA] Bat dau phan tich du lieu

[2026-11-05T00:31:35+07:00] [SUCCESS] [EDA] Hoan thanh EDA trong 0.5634 giay

[2026-11-05T00:31:35+07:00] [SUCCESS] [EDA] Hoan thanh trong 0.58 giay.

[2026-11-05T00:31:35+07:00] [START] [PREPROCESSING] Bat dau buoc PREPROCESSING.

[2026-11-05T00:31:35+07:00] [START] [PREPROCESSING] Bat dau tien xu ly du lieu

[2026-11-05T00:31:39+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh tien xu ly trong 3.9824 giay

[2026-11-05T00:31:39+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh trong 4.00 giay.

[2026-11-05T00:31:39+07:00] [START] [TRAINING] Bat dau buoc TRAINING.

[2026-11-05T00:31:39+07:00] [START] [TRAINING] Bat dau huan luyen mo hinh

[2026-11-05T00:32:11+07:00] [SUCCESS] [TRAINING - Logistic Regression] Training completed in 31.60 seconds

[2026-11-05T00:32:12+07:00] [SUCCESS] [TRAINING - Naive Bayes] Training completed in 0.37 seconds

[2026-11-05T00:32:19+07:00] [SUCCESS] [TRAINING - SVM] Training completed in 7.84 seconds

[2026-11-05T00:32:20+07:00] [SUCCESS] [TRAINING] Hoan thanh training trong 40.2721 giay

[2026-11-05T00:32:20+07:00] [SUCCESS] [TRAINING] Hoan thanh trong 40.32 giay.

[2026-11-05T00:32:20+07:00] [START] [EVALUATION] Bat dau buoc EVALUATION.

[2026-11-05T00:32:20+07:00] [START] [EVALUATION] Bat dau danh gia cac mo hinh.

[2026-11-05T00:32:20+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Train] Danh gia Logistic Regression tren Train hoan tat. F1=0.9262, AUC=0.9716

[2026-11-05T00:32:20+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Test] Danh gia Logistic Regression tren Test hoan tat. F1=0.8944, AUC=0.9597

[2026-11-05T00:32:21+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Train] Danh gia Naive Bayes tren Train hoan tat. F1=0.9938, AUC=1.0000

[2026-11-05T00:32:21+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Test] Danh gia Naive Bayes tren Test hoan tat. F1=0.9554, AUC=0.9935

[2026-11-05T00:32:21+07:00] [SUCCESS] [EVALUATION - SVM - Train] Danh gia SVM tren Train hoan tat. F1=0.1699, AUC=0.9748

[2026-11-05T00:32:21+07:00] [SUCCESS] [EVALUATION - SVM - Test] Danh gia SVM tren Test hoan tat. F1=0.1654, AUC=0.9647

[2026-11-05T00:32:23+07:00] [SUCCESS] [EVALUATION] Danh gia hoan tat trong 3.38 giay.

[2026-11-05T00:32:23+07:00] [SUCCESS] [EVALUATION] Hoan thanh trong 3.40 giay.

[2026-11-05T00:32:23+07:00] [SUCCESS] [PIPELINE] Hoan thanh toan bo quy trinh trong 48.31 giay.